

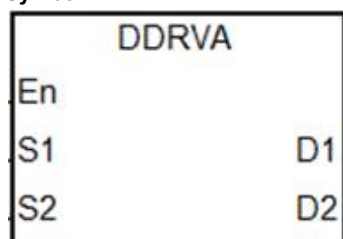
API	Instruction			Operand	Description
2707	D	DRVA		S ₁ , S ₂ , D ₁ , D ₂	Absolute position control

Device	X	Y	M	S	T	C	HC	D	FR	SM	SR	E	K	16#	"\$"	F
S_1							●	●	●		○		○	○		
S_2							●	●	●		○		○	○		
D_1		○														
D_2		○	○													

Data type	BOOL	WORD	DWORD	LWORD	UINT	INT	DINT	LINT	REAL	LREAL	TMR	CNT	STRING
S_1			●				●						
S_2			●				●						
D_1	●												
D_2	●												

Pulse Instruction	16-bit instruction	32-bit instruction
—	—	AS

Symbol



S_1 : Number of output pulses (absolute positioning)

S_2 : Pulse output frequency

D_1 : Pulse output device

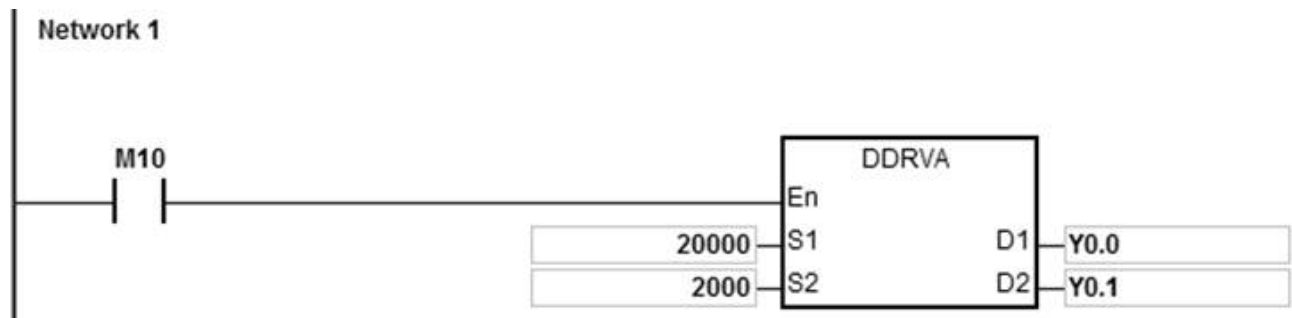
D_2 : Pulse direction output device

Explanation

- This instruction specifies the output pulse setting for absolute positioning. S_1 is the number of output pulses (absolute positioning). The range is between -2,147,483,648 to +2,147,483,647, and +/- signs indicate the positive and negative pulse directions. When the instruction is executed, it checks the current position with S_1 . If S_1 is 0, it indicates not outputting and output completion flag not be set to ON. If 0 is a possible output number in your program, it is suggested to add more conditions in your program to rule out this possibility.
- S_2 is the pulse output frequency and the range is between 0 Hz–4 MHz for the line driver output models and 0 Hz–200 kHz for the open collector output models. If the value in S_2 is less than the Start/end frequency (Hz) set in SR (refer to the SR table of DRVI instruction for more details), PLC operates according to the values set in SR.
- Absolute positioning means that the instruction outputs pulses to move from current position until the specified target position is reached. For example, the number of output pulses at current position is 100 and the number of pulses at the target position S_1 is set to 1000. So the number of the actual output pulses is 1000-100=900.
- Refer to the DDRVI instruction (API 2706) for more explanation.

Example

If the value of the present output position SR460 (32-bit) is 100 and M10 is ON, then the DDRVA instruction causes Y0.0 to output pulses at 2 kHz until the value in SR460 becomes 20,000 (absolute positioning). Y0.1 = OFF means the positive direction.



Additional remarks

1. The following graph shows absolute positioning: the way of specifying the distance from the center (zero point).

